

INTERNSHIP PROJECT REPORT

WEBSITE TRAFFIC ANALYSIS

A Data Analysis Project using Python, Pandas, NumPy and Matplotlib

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DECLARATION

I hereby declare that this report presents the work completed for the Website Traffic Analysis project. The analysis, tables, observations and recommendations are based on the supplied Jupyter Notebook and its recorded outputs. Any institutional formatting, personal details and certification statements should be completed according to the official submission requirements.

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my project supervisor, faculty members and everyone who provided guidance during the completion of this internship project. I am also thankful for the learning opportunity provided through practical work with Python, Pandas, NumPy and Matplotlib. The project helped strengthen my understanding of data preparation, exploratory analysis, visualization and communicating analytical findings.

ABSTRACT

This project presents an exploratory analysis of website traffic event data. The supplied dataset contains 226,278 recorded events across nine columns and covers the period from 19 August 2021 to 25 August 2021. The analysis examines event distribution, daily traffic, geographic activity, content/link performance, artist and track performance, and an event-level engagement measure. The notebook identifies 212 countries, 11,993 cities, 2,420 artists, 3,563 tracks and 3,839 links. Pageviews account for 62.76% of events, clicks for 24.63%, and previews for 12.61%, resulting in an event-level engagement rate of 37.24% as defined in the notebook. The analysis also records a substantial number of exact duplicate rows, while noting that the available fields are insufficient for reliable user-level or session-level metrics such as bounce rate and session duration.

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1. INTRODUCTION

Website traffic analysis is the process of examining recorded interactions with a website or digital content in order to understand activity patterns, audience distribution and content performance. In this project, the available data is event-oriented rather than session-oriented. Each record describes an observed event together with date, geographic information and content-related attributes.

The supplied notebook uses Python-based data analysis to transform raw event records into summaries and visualizations. The work is intentionally limited to metrics that can be supported by the available columns. This is important because the dataset does not contain user identifiers, session identifiers, timestamps, page URLs or referral information.

2. PROJECT OBJECTIVES

- Analyze website event types and overall traffic activity.
- Identify changes in traffic across the seven-day period.
- Identify high-activity countries and cities.
- Evaluate high-volume content links.
- Analyze artist and track event volume.
- Measure an available event-level engagement pattern.
- Document data-quality issues such as missing values and duplicate rows.
- Provide business-oriented recommendations supported by the observed data.
- Clearly document which requested user/session metrics cannot be calculated from the available fields.

3. TOOLS AND TECHNOLOGIES

Tool / Library	Purpose
Python	Core programming and analysis environment
Jupyter Notebook	Interactive development and presentation
Pandas	Data loading, cleaning, grouping and aggregation
NumPy	Numerical/data-analysis support
Matplotlib	Charts and visualizations
CSV dataset	Source data loaded as traffic.csv

The notebook imports Pandas, NumPy and Matplotlib and sets display options before loading the traffic.csv dataset. The analysis is performed using dataframe operations, grouping, aggregation and Matplotlib visualizations.

4. DATASET DESCRIPTION

The dataset contains 226,278 rows and 9 columns. The available fields are event, date, country, city, artist, album, track, isrc and linkid. The recorded period extends from 19 August 2021 through 25 August 2021.

Field	Description based on notebook usage
event	Recorded activity type: pageview, click or preview
date	Calendar date of the recorded event
country	Country associated with the event

city	City associated with the event
artist	Artist associated with the content
album	Album associated with the content
track	Track associated with the content
isrc	ISRC identifier; some values are missing
linkid	Identifier used by the notebook to group content links

	event	date	country	city	artist	album	track	isrc	linkid
0	click	2021-08-21	Saudi Arabia	Jeddah	Tesher	Jalebi Baby	Jalebi Baby	QZNWQ2070741	2d896d31-97b6-4869-967b-1c5fb9cd4bb8
1	click	2021-08-21	Saudi Arabia	Jeddah	Tesher	Jalebi Baby	Jalebi Baby	QZNWQ2070741	2d896d31-97b6-4869-967b-1c5fb9cd4bb8
2	click	2021-08-21	India	Ludhiana	Reyanna Maria	So Pretty	So Pretty	USUM72100871	23199824-9cf5-4b98-942a-34965c3b0cc2
3	click	2021-08-21	France	Unknown	Simone & Simaria, Sebastian Yatra	No Llores Más	No Llores Más	BRUM72003904	35573248-4e49-47c7-af80-08a960fa74cd
4	click	2021-08-21	Maldives	Malé	Tesher	Jalebi Baby	Jalebi Baby	QZNWQ2070741	2d896d31-97b6-4869-967b-1c5fb9cd4bb8

5. DATA LOADING AND INITIAL INSPECTION

The dataset is loaded with Pandas using `read_csv("traffic.csv")`. The notebook confirms successful loading and reports 226,278 rows and 9 columns. Initial inspection displays the dataframe shape, column names, first five rows and data types.

At the raw inspection stage, all nine columns are represented as object/string-like data. The date column is subsequently converted into datetime format during cleaning.

Initial metric	Recorded value
Rows	226,278
Columns	9
Date range	19 Aug 2021 – 25 Aug 2021
Event types	click, preview, pageview
Unique countries	212
Unique cities	11,993
Unique artists	2,420

Unique tracks	3,563
Unique links	3,839

6. DATA QUALITY ASSESSMENT

The notebook performs a missing-value check, duplicate-row check, date-range check and event-type check. Missing values are relatively small for most fields, but the ISRC field has 7,121 missing values, corresponding to 3.15% of the dataset.

Column	Missing values	Missing percentage
event	0	0.00%
date	0	0.00%
country	11	0.00%
city	11	0.00%
artist	37	0.02%
album	5	0.00%
track	5	0.00%
isrc	7121	3.15%
linkid	0	0.00%

A major data-quality observation is the presence of 103,711 exact duplicate rows, representing 45.83% of the dataset. The notebook deliberately retains these rows because the available information does not provide enough evidence to determine whether repeated records are genuine repeated events or duplicate ingestion.

	Missing Values	Missing Percentage
event	0	0.00
date	0	0.00
country	11	0.00
city	11	0.00
artist	37	0.02
album	5	0.00
track	5	0.00
isrc	7121	3.15
linkid	0	0.00

Exact Duplicate Rows: 103,711
Duplicate Percentage: 45.83%

Date Range:
Start: 2021-08-19
End: 2021-08-25

Unique Event Types:
['click' 'preview' 'pageview']

7. DATA CLEANING AND PREPARATION

The cleaning process converts the date field to datetime using error coercion, strips leading/trailing whitespace from text columns, converts event values to lowercase and fills missing categorical values with “Unknown”. After cleaning, the notebook reports zero invalid dates and zero remaining missing values.

The notebook does not remove exact duplicate rows. This is an explicit analytical decision and should be stated in the final submission rather than implying that the dataset was fully deduplicated.

- Convert date to datetime.
- Trim whitespace in text columns.
- Standardize event names to lowercase.
- Replace missing categorical values with “Unknown”.
- Retain exact duplicates because event-level uniqueness cannot be established from the available fields.

8. EXPLORATORY DATA ANALYSIS

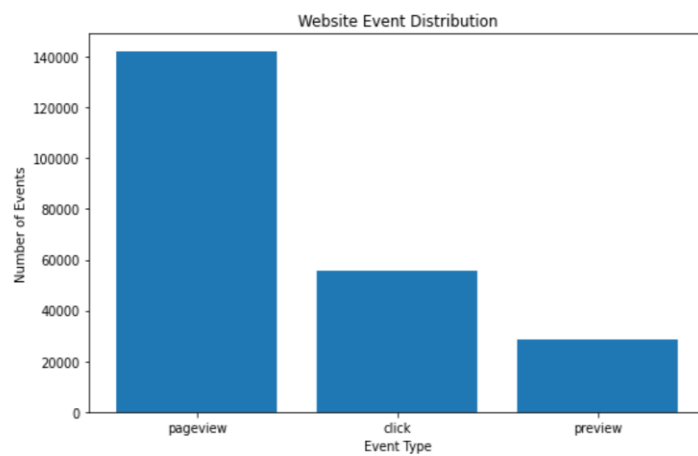
Exploratory Data Analysis (EDA) is used to summarize the dataset and identify patterns before drawing conclusions. The project examines event composition, time-based activity, geography, content links, artists, tracks and engagement.

9. EVENT ANALYSIS

The dataset contains three recorded event types: pageview, click and preview. Pageviews dominate the event distribution with 142,015 records (62.76%). Clicks account for 55,732 records (24.63%), while previews account for 28,531 records (12.61%).

Event	Count	Percentage
Pageview	142015	62.76%
Click	55732	24.63%
Preview	28531	12.61%

The notebook defines event-level engagement as (clicks + previews) / total events. Using this definition, the engagement rate is 37.24%. This should be described as an event-level measure rather than a user engagement rate.

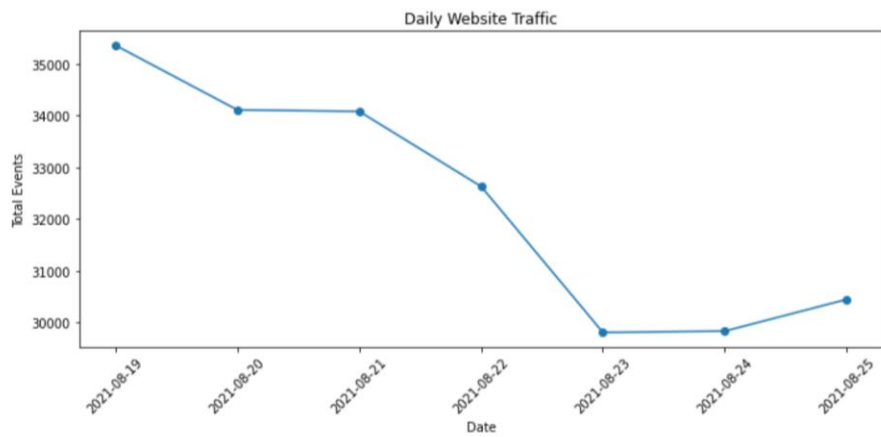


10. DAILY TRAFFIC ANALYSIS

Daily traffic analysis summarizes event activity across the seven dates covered by the dataset. Total events decrease from the first day to the lowest point on 23 August, followed by a small recovery on 24 and 25 August.

Date	Clicks	Pageviews	Previews	Total events
19 Aug 2021	9207	22366	3788	35361
20 Aug 2021	8508	21382	4222	34112
21 Aug 2021	8071	21349	4663	34083
22 Aug 2021	7854	20430	4349	32633
23 Aug 2021	7315	18646	3847	29808
24 Aug 2021	7301	18693	3840	29834
25 Aug 2021	7476	19149	3822	30447

The peak traffic day is 19 August 2021 with 35,361 events. The lowest traffic day is 23 August 2021 with 29,808 events.



11. GEOGRAPHIC ANALYSIS

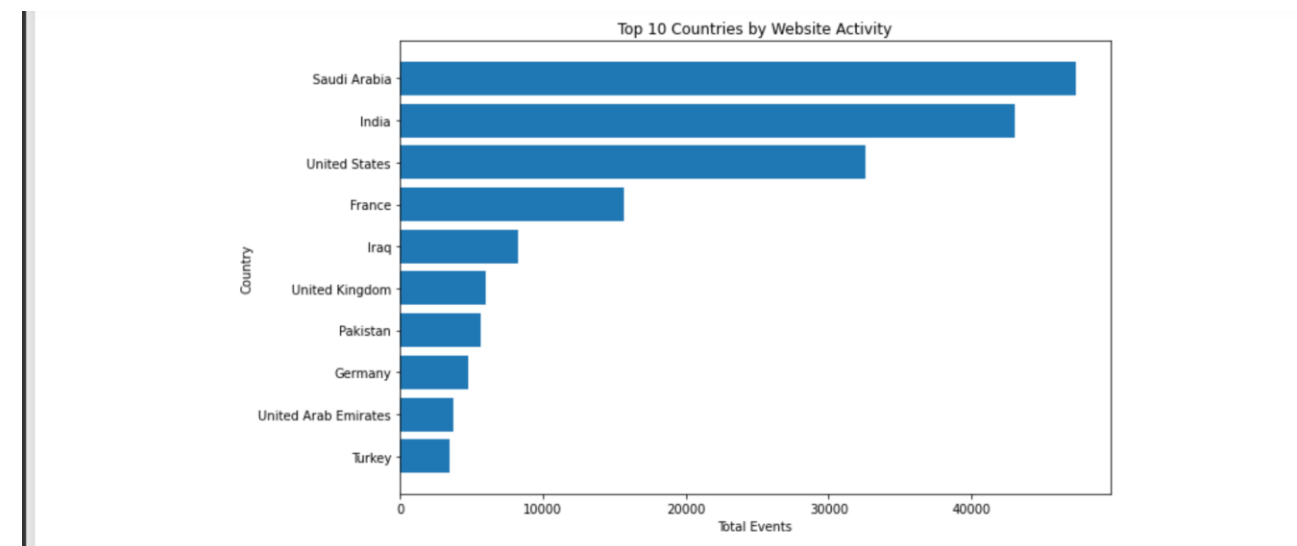
Geographic analysis groups events by country and city. The notebook explicitly notes that these are event counts and should not be interpreted as unique-user counts.

Country	Total events	Pageviews	Clicks	Previews
Saudi Arabia	47334	28873	9495	8966
India	42992	27286	11609	4097
United States	32558	20839	9533	2186
France	15661	9674	4054	1933
Iraq	8260	4897	1836	1527
United Kingdom	5970	3845	1517	608
Pakistan	5644	3212	1653	779
Germany	4794	3141	1225	428
United Arab Emirates	3702	2335	766	601
Turkey	3514	2462	654	398

Saudi Arabia has the highest recorded activity with 47,334 events, followed by India with 42,992 and the United States with 32,558. At city level, Jeddah has 22,791 events and Riyadh has 17,454. The presence of an “Unknown” city category is also notable.

City	Total events
Jeddah	22791

Riyadh	17454
Unknown	16672
Hyderabad	3682
Dammam	3276
Erbil	3105
Kuwait City	3010
Lucknow	2808
Muscat	2717
Delhi	2567



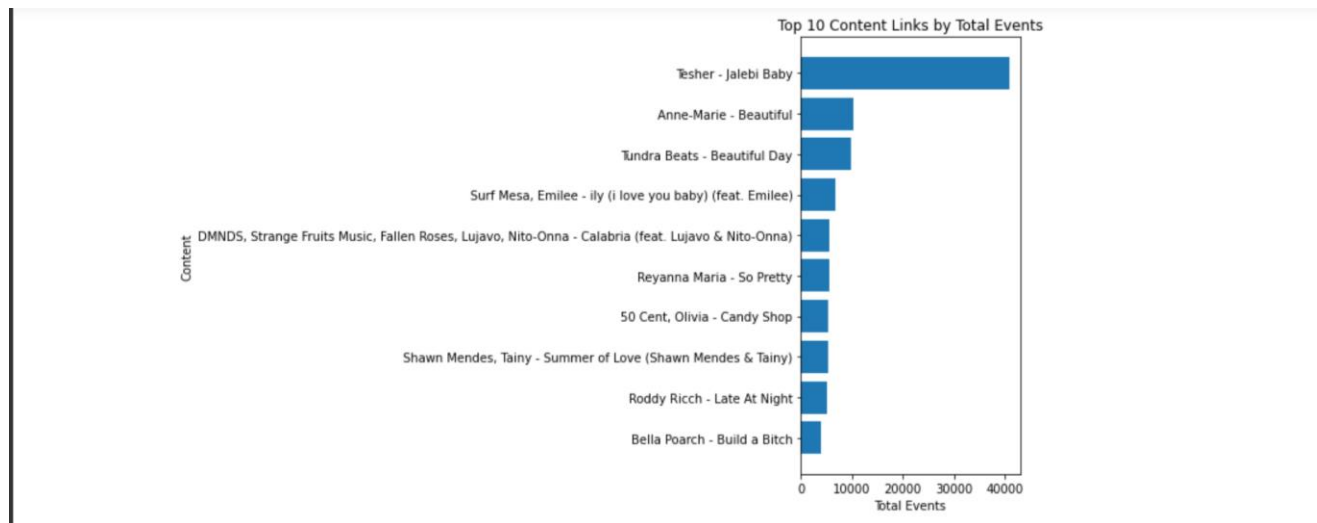
12. CONTENT AND LINK PERFORMANCE

The linkid field is used to group content and evaluate which links receive the highest volume of recorded activity. The top content link is associated with Teshar and the track “Jalebi Baby”, with 40,841 total events.

Artist	Track	Total events	Pageviews	Clicks	Previews
Teshar	Jalebi Baby	40841	25175	9692	5974
Anne-Marie	Beautiful	10314	6600	2109	1605

Tundra Beats	Beautiful Day	9750	5981	2198	1571
Surf Mesa, Emilee	ily (i love you baby) (feat. Emilee)	6733	4303	1429	1001
DMNDS / collaborators	Calabria (feat. Lujavo & Nito-Onna)	5512	3354	1500	658
Reyanna Maria	So Pretty	5437	3532	1187	718
50 Cent, Olivia	Candy Shop	5367	3407	1111	849
Shawn Mendes, Tainy	Summer of Love	5352	3299	1353	700
Roddy Ricch	Late At Night	5036	3039	1364	633
Bella Poarch	Build a Bitch	3879	2440	827	612

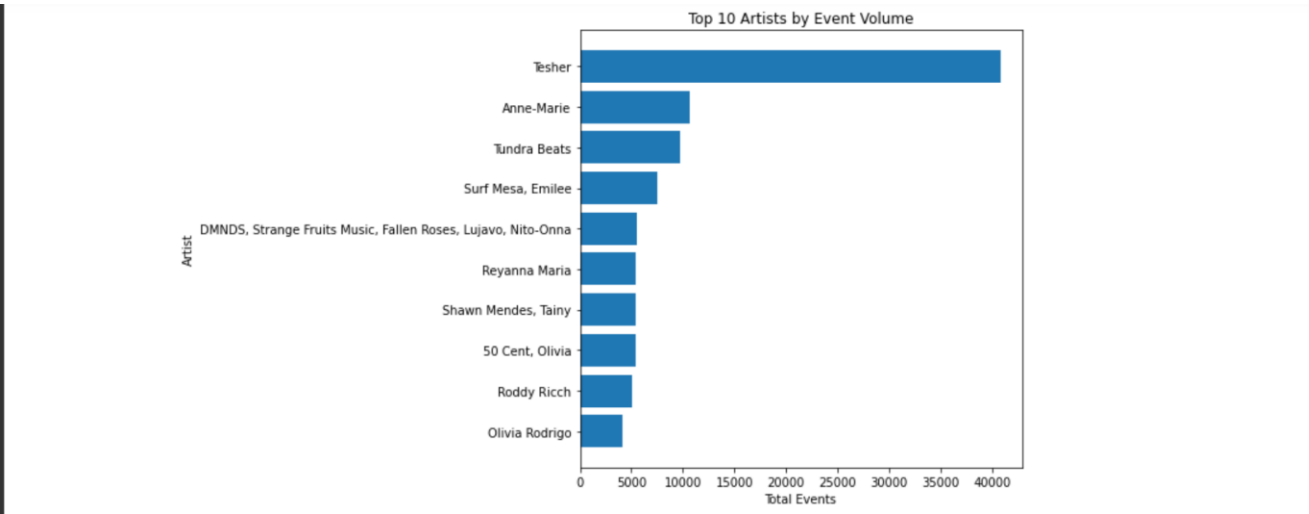
The ranking indicates a strong concentration of activity around the leading content item. The first link has substantially more recorded events than the second-ranked link. This makes high-volume content a useful candidate for promotion, placement experiments and further investigation.



13. ARTIST AND TRACK PERFORMANCE

Artist and track summaries are generated by counting the number of records associated with each artist and track. The notebook identifies Teshar as the highest-volume artist and “Jalebi Baby” as the highest-volume track.

The analysis is based on event volume. It therefore indicates which artists/tracks are associated with the most recorded activity, but it does not establish unique listeners, conversion rates, retention or satisfaction.



14. ENGAGEMENT ANALYSIS

Because user and session identifiers are unavailable, the notebook measures engagement at the event level. For each link, the engagement rate is calculated as (clicks + previews) / total events. A minimum threshold of 100 total events is used before ranking high-engagement links, reducing the influence of extremely small samples.

This approach provides a useful descriptive comparison of event composition, but it should not be presented as a true user engagement or conversion metric.

15. SESSION AND USER-LEVEL ANALYSIS LIMITATION

The internship task references metrics such as sessions, users, bounce rate, average session duration, referral sources, user journeys, entry pages and exit pages. The supplied dataset does not contain the fields required to calculate these metrics reliably.

Requested metric	Why it cannot be reliably calculated
Users	No user ID
Sessions	No session ID or timestamp-level event sequence
Bounce rate	No reliable session boundaries or page/session definition
Average session duration	No event timestamps or session start/end

Referral sources	No referral-source field
User journeys	No user/session identifiers and no event timestamps
Entry / exit pages	No page URL/page identifier and no session sequence
Conversions	No explicit conversion event/field

The correct analytical approach is to document this limitation rather than inventing assumptions. The notebook follows this principle.

16. KEY FINDINGS

- 226,278 website events are available across nine fields.
- Three event types are present: pageview, click and preview.
- Pageviews represent 62.76% of recorded events.
- Clicks and previews together represent an event-level engagement rate of 37.24%.
- Saudi Arabia is the highest-activity country, followed by India and the United States.
- Jeddah and Riyadh are the two highest-activity named cities in the city summary.
- 19 August 2021 is the peak traffic day; 23 August 2021 is the lowest.
- Teshar and “Jalebi Baby” are the highest-volume artist and track identified in the notebook.
- The dataset contains 103,711 exact duplicate rows (45.83%).
- User-level and session-level metrics cannot be reliably calculated with the supplied fields.

17. BUSINESS RECOMMENDATIONS

17.1 Prioritize High-Performing Content

Promote high-volume and high-engagement content in prominent website positions. The leading content link, associated with Teshar and “Jalebi Baby”, can be considered for additional placement or recommendation testing.

17.2 Focus on High-Activity Markets

Major markets such as Saudi Arabia, India and the United States account for substantial event activity. Localized content, campaigns or audience-specific recommendations can be explored.

17.3 Improve Content Recommendations

Popular artists and tracks can be used to design recommendation modules intended to encourage further exploration. Any impact should be evaluated using controlled experiments.

17.4 Improve Website Tracking

Future data collection should include user IDs, session IDs, event timestamps, page URLs, referral sources and explicit conversion events. These fields would enable deeper funnel and journey analysis.

17.5 Conduct A/B Testing

Website layouts, calls-to-action, content placement and recommendation strategies can be tested systematically. Future experiments should define a measurable success metric before implementation.

18. PROJECT LIMITATIONS

- The dataset covers only seven calendar days.
- Event records do not contain user IDs or session IDs.
- Event timestamps are not available in the supplied fields.
- Page URLs and referral sources are unavailable.
- Conversion events are unavailable.
- Exact duplicate records are retained because their validity cannot be determined from the available fields.
- Event counts should not be interpreted as unique-user counts.
- Engagement is an event-level descriptive measure, not a session/user engagement metric.
- Some geographic/content fields contain missing values that are converted to “Unknown”.

19. FUTURE SCOPE

Future versions of the project can extend the analysis once richer tracking data is available. Possible additions include sessionization, unique-user estimation, funnel analysis, bounce-rate calculation, session duration, referral-source analysis, landing/exit-page analysis, cohort analysis, conversion tracking and dashboard development.

- Add event timestamps and session identifiers.
- Add anonymized user identifiers where appropriate.
- Capture page URL and page type.
- Capture traffic source and campaign information.
- Add conversion and revenue-related events if applicable.
- Create a Power BI/Tableau dashboard for management reporting.
- Monitor the same metrics over a longer period to identify weekly and monthly patterns.

20. CONCLUSION

The Website Traffic Analysis project demonstrates a complete introductory data-analysis workflow using Python. The notebook loads and inspects the dataset, checks data quality, cleans fields, summarizes event activity, analyzes daily traffic, identifies geographic and content patterns, measures event-level engagement and develops business recommendations.

The analysis provides clear evidence of high activity in Saudi Arabia, India and the United States, a dominant share of pageviews, strong activity around selected content, and meaningful variation in traffic across the seven-day period. At the same time, the project appropriately recognizes the limits of the data: user and session metrics cannot be calculated reliably without additional tracking fields.

The project therefore provides both analytical findings and a practical roadmap for improving future data collection and website measurement.